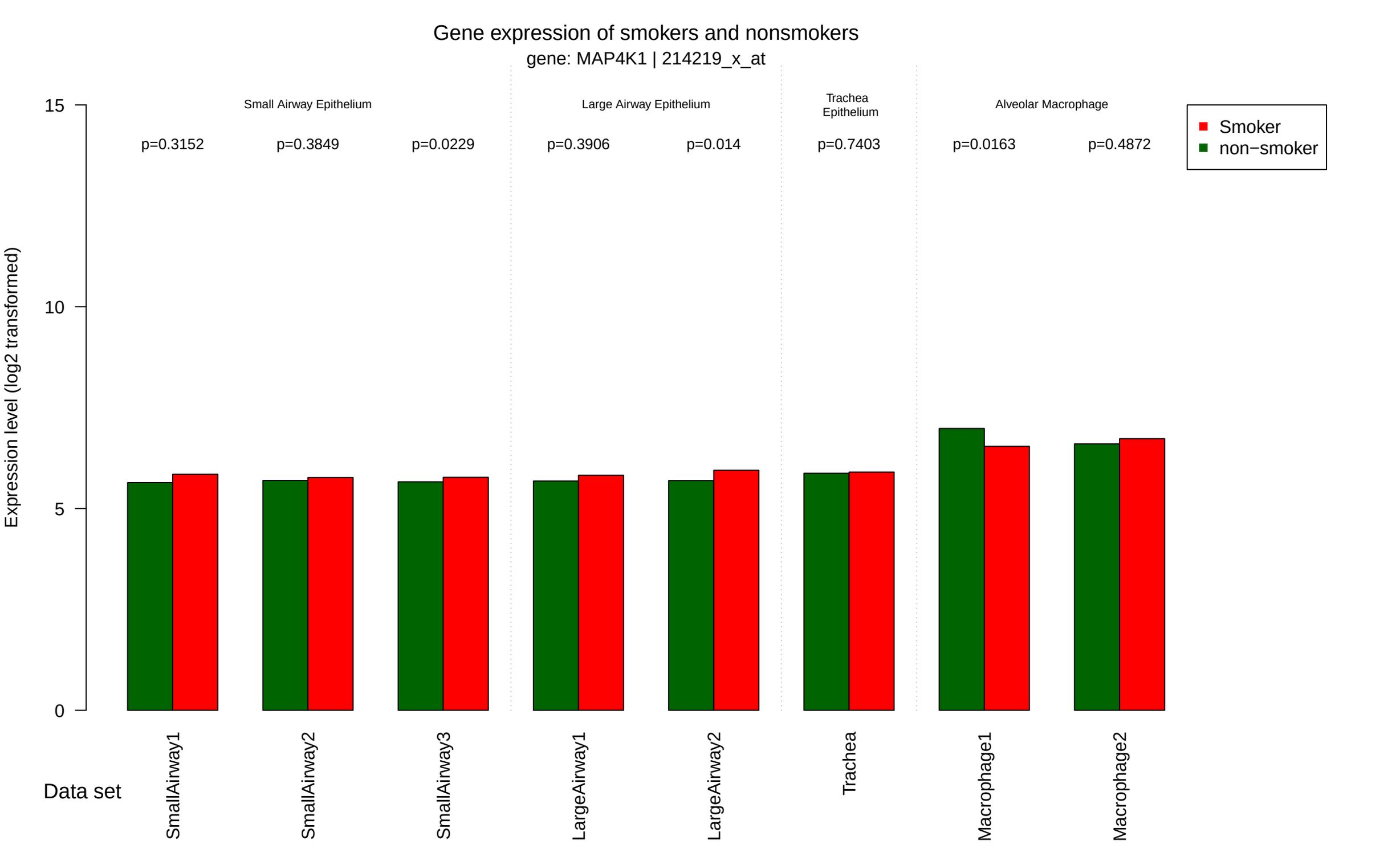
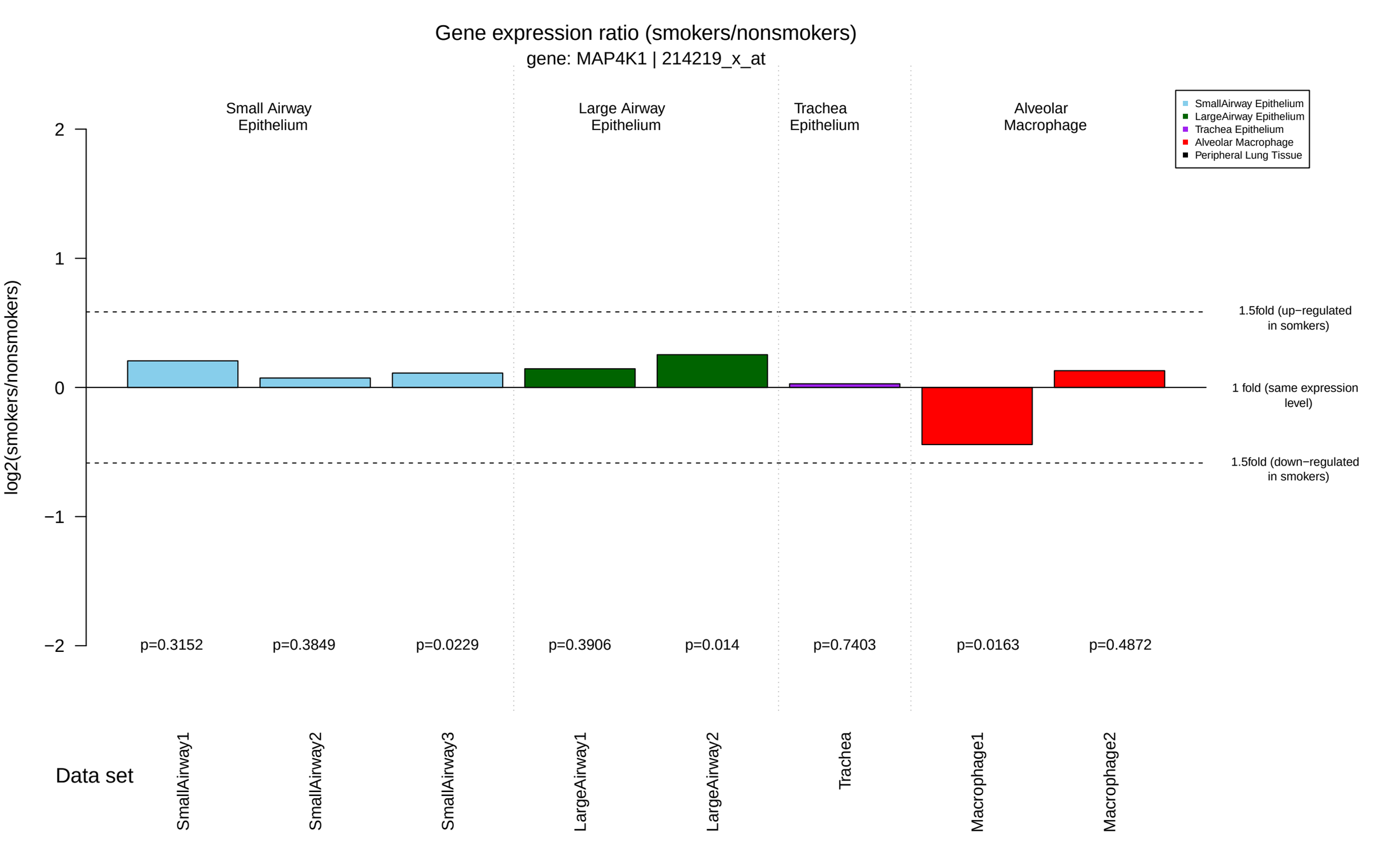
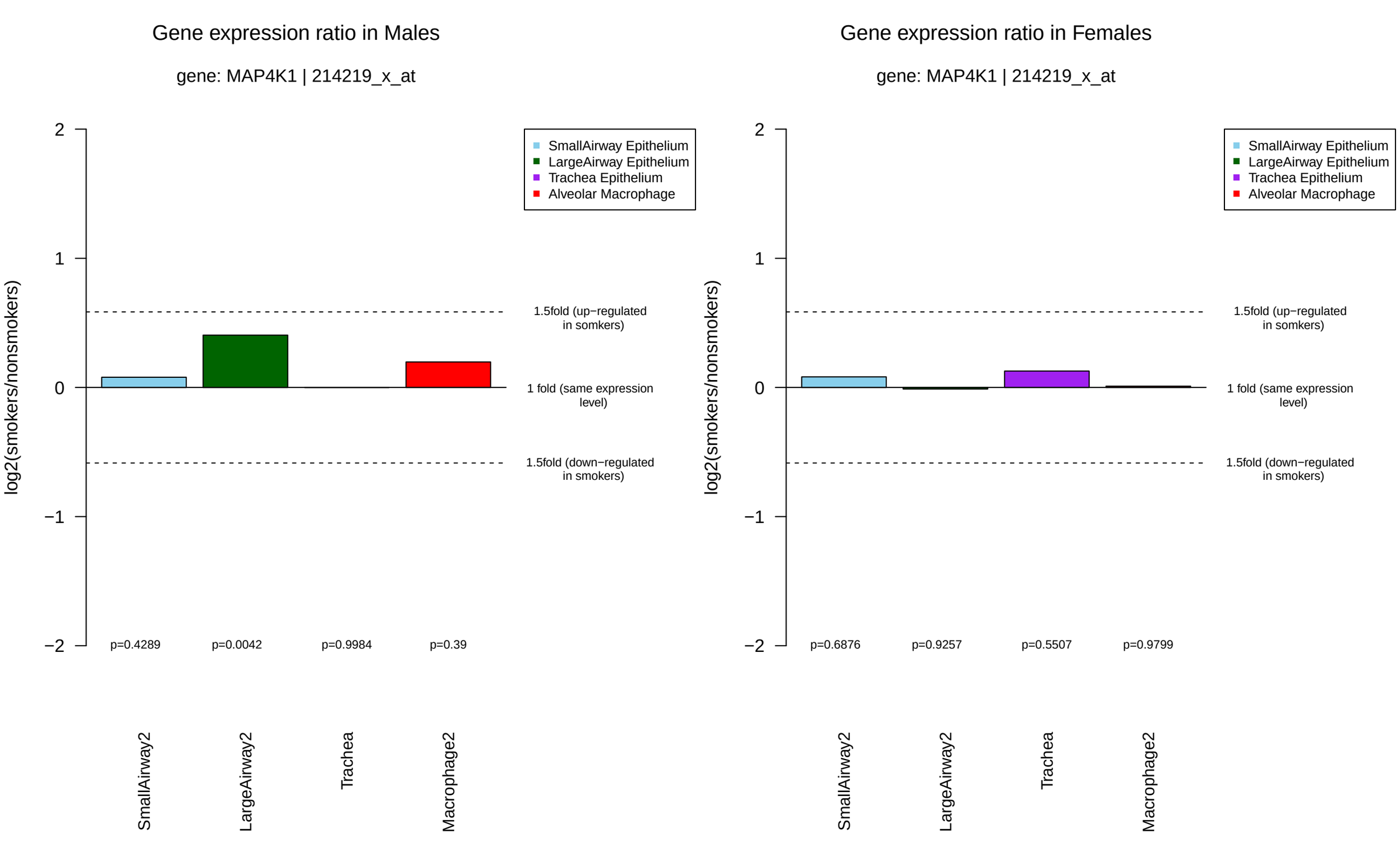
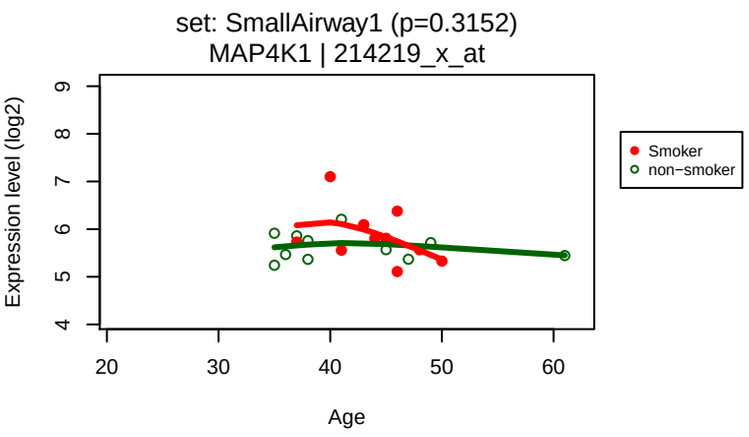
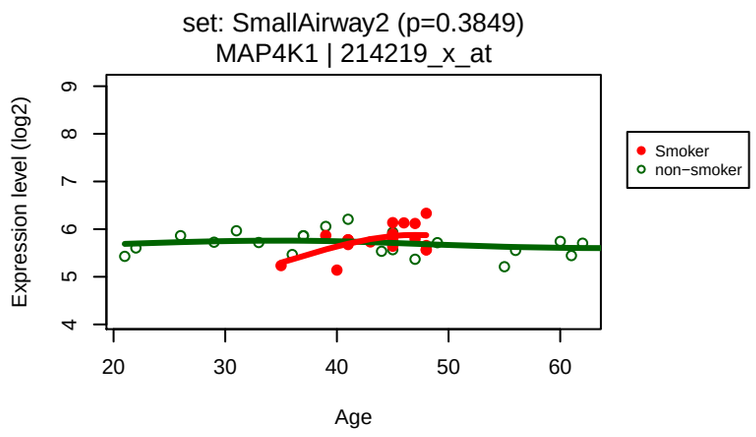
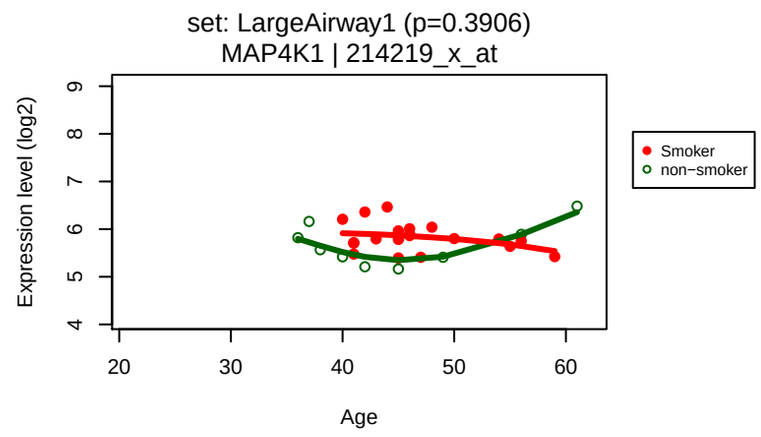

Gene expression of smokers and nonsmokers
gene: MAP4K1 | 214219_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3152
p=0.3849
p=0.0229
p=0.3906
p=0.014
p=0.7403
p=0.0163
p=0.4872
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

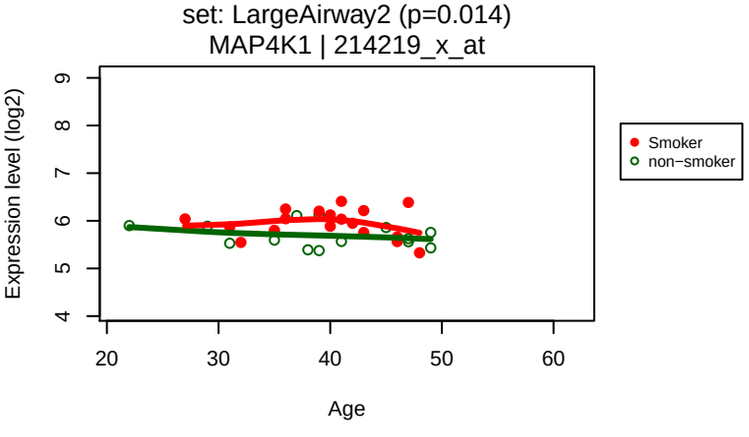

Gene expression ratio (smokers/nonsmokers)
gene: MAP4K1 | 214219_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3152
p=0.3849
p=0.0229
p=0.3906
p=0.014
p=0.7403
p=0.0163
p=0.4872
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

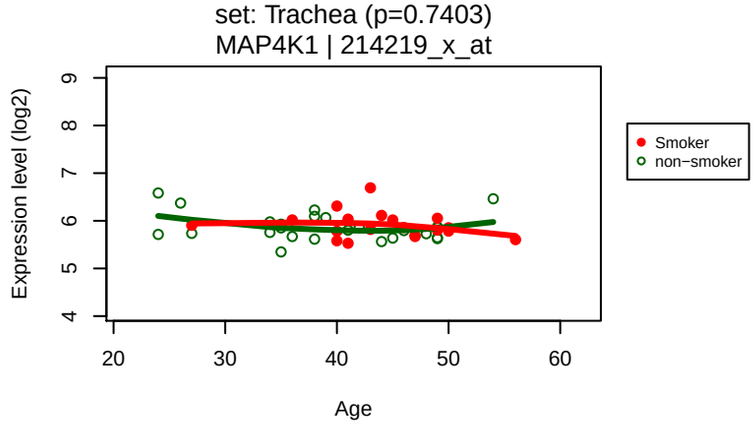

Gene expression ratio in Males
gene: MAP4K1 | 214219_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4289
p=0.0042
p=0.9984
p=0.39
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAP4K1 | 214219_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6876
p=0.9257
p=0.5507
p=0.9799
SmallAirway2
LargeAirway2
Trachea
Macrophage2

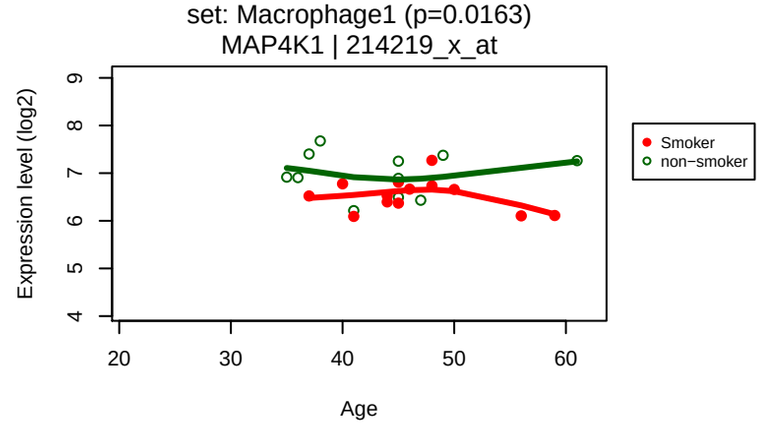

set: SmallAirway1 (p=0.3152)
MAP4K1 | 214219_x_at
Expression level (log2)
Age
Smoker
non−smoker

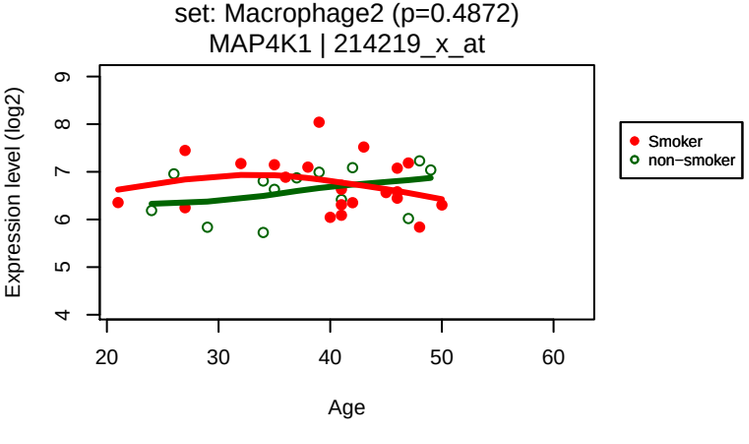

set: SmallAirway2 (p=0.3849)
MAP4K1 | 214219_x_at
Expression level (log2)
Age
Smoker
non−smoker

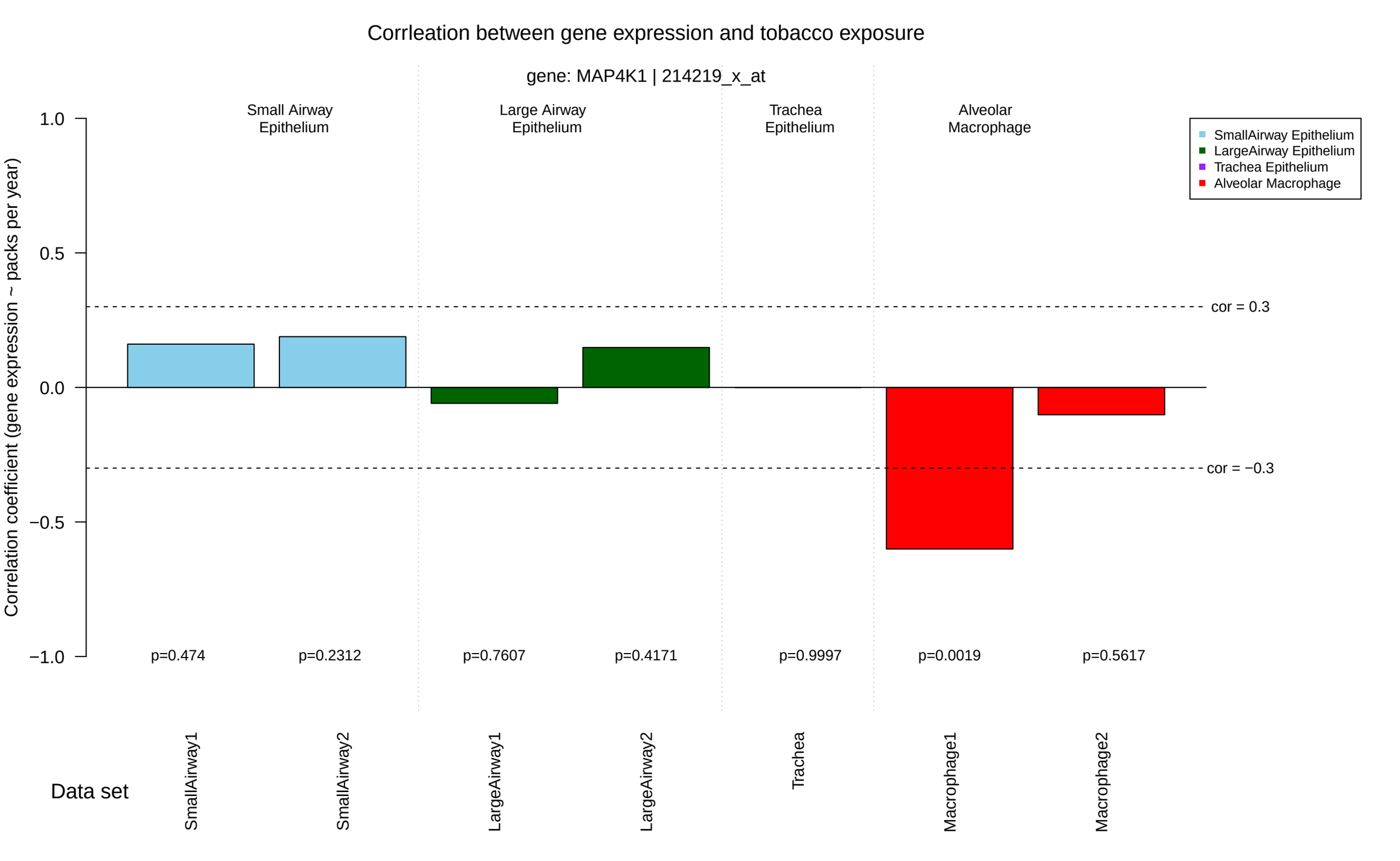

set: LargeAirway1 (p=0.3906)
MAP4K1 | 214219_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.014)
MAP4K1 | 214219_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7403)
MAP4K1 | 214219_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0163)
MAP4K1 | 214219_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4872)
MAP4K1 | 214219_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAP4K1 | 214219_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.474
p=0.2312
p=0.7607
p=0.4171
p=0.9997
p=0.0019
p=0.5617
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2